Introduction - Homework 2

- Backwards Heat Equation
  - Bayesian Framework
  - Reduced Observational Region
- Abel Problem and Generalised Tikhonov
  - Uniqueness and Existence
  - Morozov Discrepancy Principle
- Feedback received on report
  - Use *augmented least-squares* for improved stability (Tikhonov)
  - Discuss null-space of central difference matrix

Backwards Heat Equation

We seek to model the 1D backwards heat equation

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0, \qquad 0 < x < \pi, \quad t \in (0, T]$$

$$u(0,t) = u(\pi,t) = 0, \qquad u(x,0) = f(x).$$

Which we model using a forward operator K such that $y(t) = Kx(t)$, where K may be discretised on $n = 128$ grid points as $A \in \mathbb{R}^{n \times n}$.

We consider the problem using a *Bayesian framework*, $Y = Ax + E$ where x is a realisation of $X \sim \mathcal{N}(0, \gamma^2 I)$ with $\gamma = 10^{-1}$ and the noise enters as $E \sim \mathcal{N}(0, \sigma^2 I), \quad \sigma = 10^{-3}$. Thus, an observation is affected by noise as $y_{\text{obs}} = y + e$

Backwards Heat Equation — Probabilistic Formulation

We obtain:

$$\pi_{\text{pr}}(x) \propto \exp\left(-\frac{x^2}{2\gamma^2}\right)$$

$$\pi(y_{\text{obs}}|x) \propto \exp\left(-\frac{(y_{\text{obs}} - Ax)^2}{2\sigma^2}\right)$$

$$\pi(y_{\text{obs}}) = \int \pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)\,\mathrm{d}x \propto 1$$

Which using *Bayes' theorem* gives the posterior distribution as

$$\pi_{\text{post}} = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y_{\text{obs}})} \propto \exp\left(-\frac{1}{2}\left(\frac{\|y_{\text{obs}} - Ax\|_2^2}{\sigma^2} + \frac{\|x\|_2^2}{\gamma^2}\right)\right)$$

$$\propto \exp\left(-\frac{1}{2}(x - \bar{x})^{\mathsf{T}}\Gamma_{\text{post}}^{-1}(x - \bar{x})\right)$$

Thus the posterior is also Gaussian with mean $\bar{x}$ and covariance Γ_{post}.

Backwards Heat Equation — Posterior Mean and Covariance

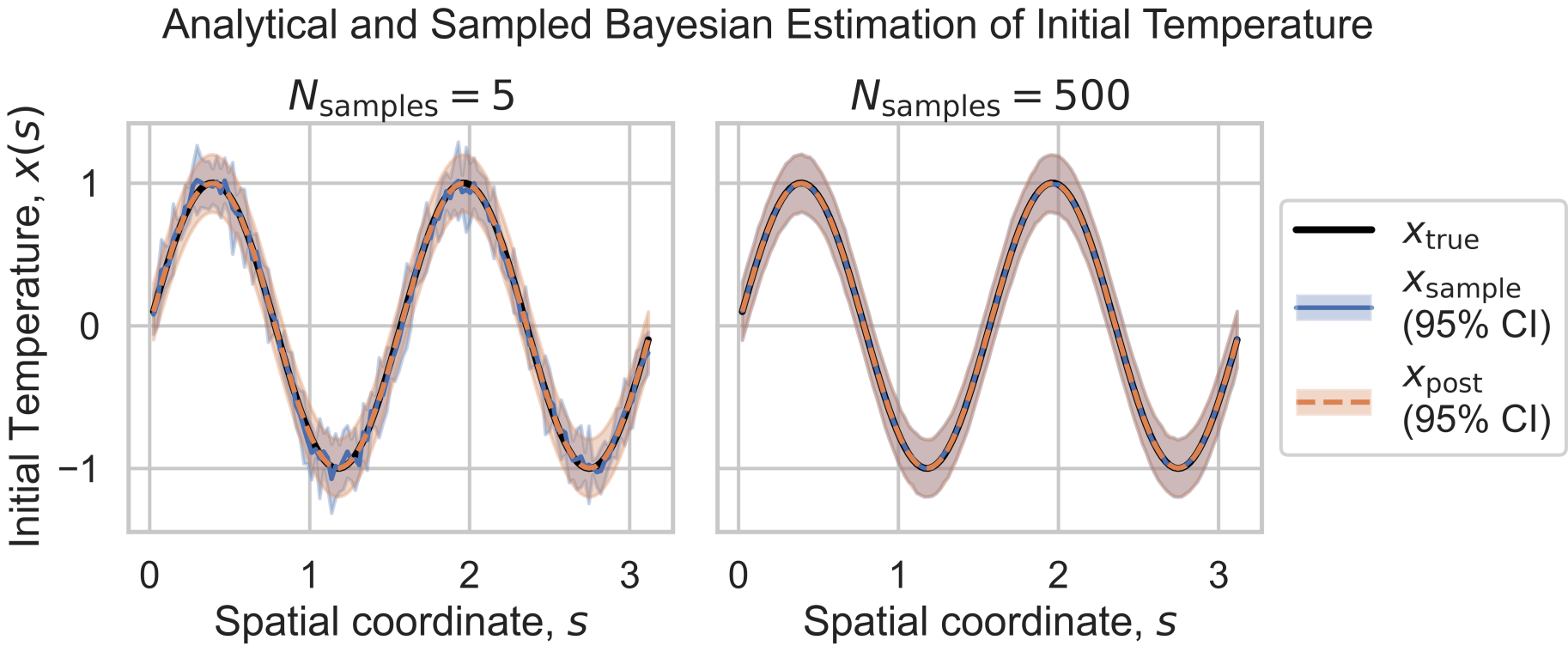
Algebraically arduous to find closed-form expressions for $\bar{x}$ and Γ_{post}, outlined as:

- Expand red expression on last slide and compare against blue expression
- Quadratic part becomes inverse covariance: $\Gamma_{\text{post}}^{-1} = \frac{A^\mathsf{T}A}{\sigma^2} + \frac{I}{\gamma^2}$
- Rest gives mean: $\bar{x} = \Gamma_{\text{post}} A^\mathsf{T} \frac{y_{\text{obs}}}{\sigma^2}$
- Obtain Γ_{post} by inverting $\Gamma_{\text{post}}^{-1}$ using *Woodbury matrix identity*:
  - $\Gamma_{\text{post}} = \gamma^2 I - \gamma^4 A^\mathsf{T}\left(\sigma^2 + \gamma^2 AA^\mathsf{T}\right)^{-1} A$
  - $\bar{x} = \gamma^2 A^\mathsf{T}\left(\sigma^2 I + \gamma^2 AA^\mathsf{T}\right)^{-1} y_{\text{obs}}$
  - Note: Can recover normal form Tikhonov by letting $\alpha = \sigma^2/\gamma^2$ and algebra
  - $\bar{x} = \left(A^\mathsf{T}A + \alpha I\right)^{-1} A^\mathsf{T} y_{\text{obs}}$

Can also prove that incorporation of observations reduces uncertainty, $\Gamma_{\text{post}} < \gamma^2 I$

- Note purple term is *symmetric* and show *positive definiteness* by considering quadratic form and transformations $z = Ay$, $B = \left(\sigma^2 + \gamma^2 AA^\mathsf{T}\right)$
- We find that B is positive definite $\Rightarrow$ B^{-1} is positive definite $\Rightarrow$ $\Gamma_{\text{post}} < \gamma^2 I$ ■

Backwards Heat Equation — Monte Carlo Sampling

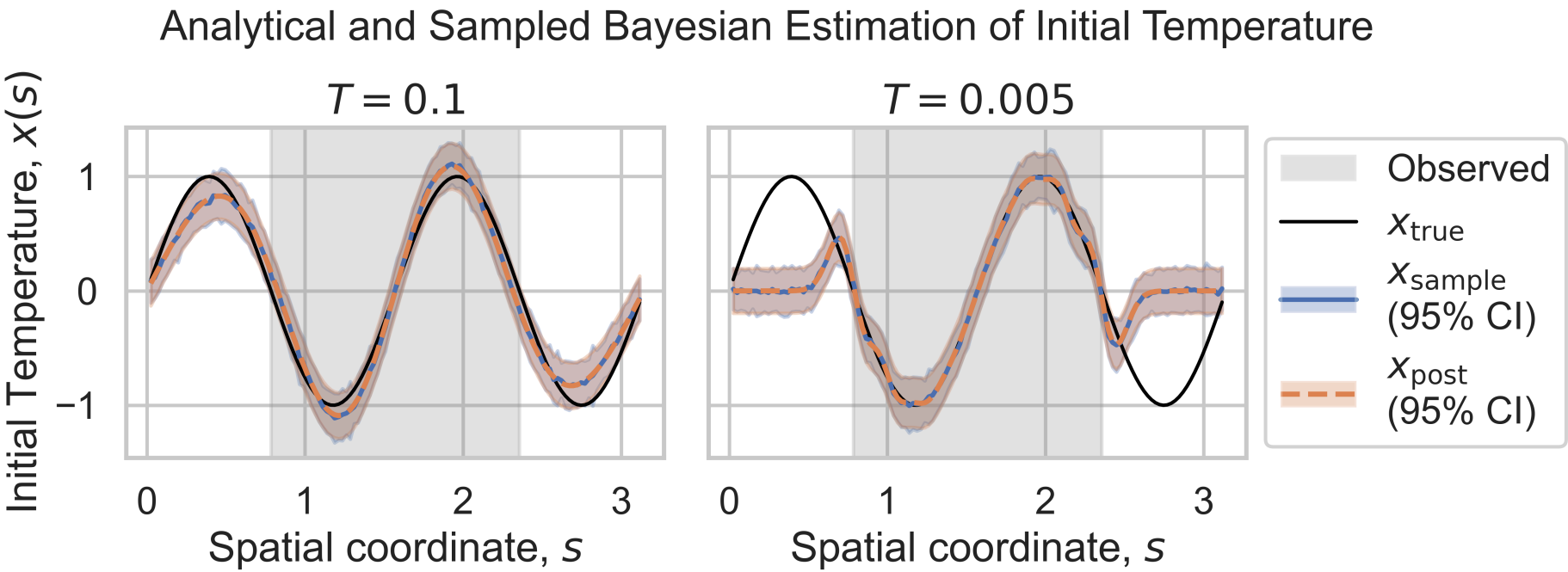


- For $N_{\text{samples}} \gg 1$ we get very good approximation of expected posterior distribution
- Mean estimates surprisingly good even for low N_{samples}

Backwards Heat Equation — Reduced Observational Region

We reduce our observational region to $s \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ instead of full $[0, \pi]$.

- Unit diffusivity, terminal time T determines ill-posedness
- Information propagates between observed and unobserved regions with time
- Smaller $T \Rightarrow$ less propagation $\Rightarrow$ harder to infer unobserved region (falls back to prior)



Abel Problem

We consider Abel problem with integral operator K such that $y(t) = Kx(t)$:

$$Kx(t) = \int_0^1 k(t,s)x(s)\,\mathrm{d}s \qquad k(t,s) = \begin{cases} \frac{1}{\sqrt{t-s}} & s < t \\ 0 & s \geq t \end{cases} \qquad x(s) = s^{3/4} - 1$$

which is discretised into $A \in \mathbb{R}^{N\times N}$ where $N = 1024$.

We introduce measurement noise of 10% relative to norm of y_{obs} to obtain $y^\delta = Ax + n$.

We find $x(s) = s^{3/4} - 1 \in H^1(0,1)$

- $x(s) \in L^2(0,1) \quad \wedge \quad x'(s) \in L^2(0,1) \quad \Rightarrow \quad x(s) \in H^1(0,1)$
- Weak derivative is just derivative since x is smooth
- Square integrals fairly trivial to compute

Abel Problem — Generalised Tikhonov Regularisation

We seek to prove existence of uniqueness of solution to generalised Tikhonov problem

$$x_\alpha = \underset{x \in H^1(0,1)}{\arg\min} \left\| Ax - y^\delta \right\|_2^2 + \alpha(\|x\|_2^2 + \|Dx\|_2^2)$$

Existence:

- *Weierstrass Extreme Value Theorem*: continuous function on non-empty compact set attains global extrema on set
- Continous by inspection
- Sublevel sets are bounded by coercivity of objective, i.e. $f(x) \to +\infty$ as $\|x\|_2 \to +\infty$
  - Derivative and residual terms are non-negative by construction
  - $\alpha\|x\|_2^2$ is trivially coercive for $\alpha > 0$

Uniqueness:

- Convexity analysis (02611 Optimisation for DS, alternatively 2nd order condition)
- Sum of convex functions is convex; norms are convex
- Non-negative scaling, raising to power $p \geq 1$ preserves convexity
- Strict convexity from Hessian: $\nabla_x^2 \|x\|_2^2 = 2I \succ 0$, thus strictly convex by second order condition

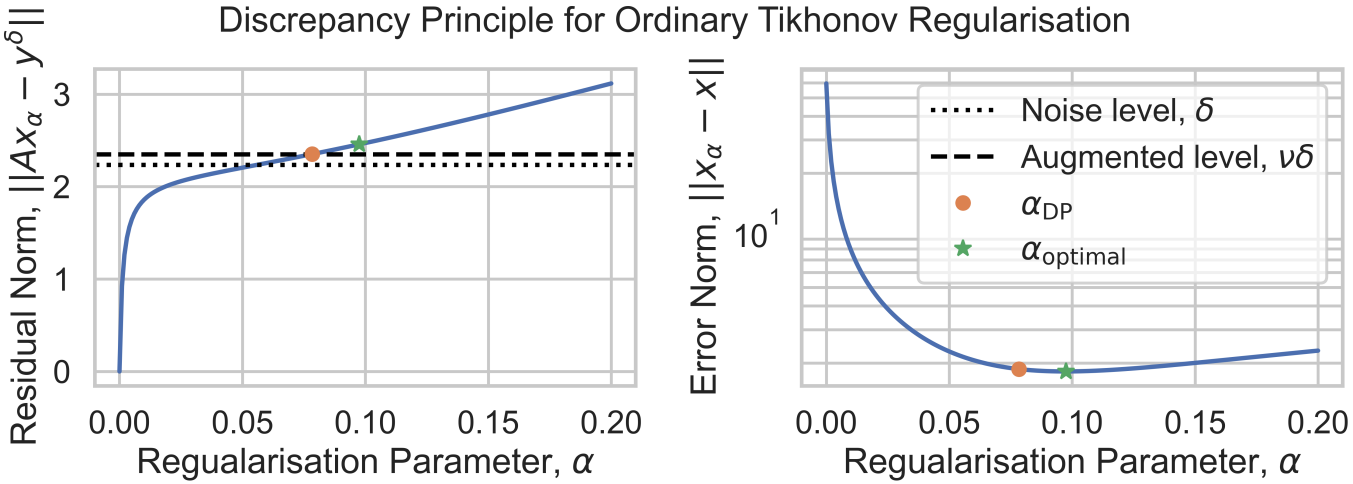
Abel Problem — Morozov Discrepancy Principle

An optimal regularisation parameter $\alpha^\star$ which minimises the reconstruction error $\|x_a - x\|_2$.
This is not available in practice as x is unknown.

Instead use *Morozov Discrepancy Principle*, which requires only estimate of noise level δ:

- Find α_{DP} such that $\left\|Kx_{\alpha_{\text{DP}}} - y^\delta\right\| \approx \nu\delta$
- $\nu = 1.05 \geq 1$ is safety factor to avoid under-regularisation
- Here found using grid search, can also be found using root-finding methods

- Slightly undershoots optimal value
- Note flatness of curve near minimum
  - This is problem dependent
  - Ill-posed = very flat



Abel Problem — MDP on Generalised Tikhonov

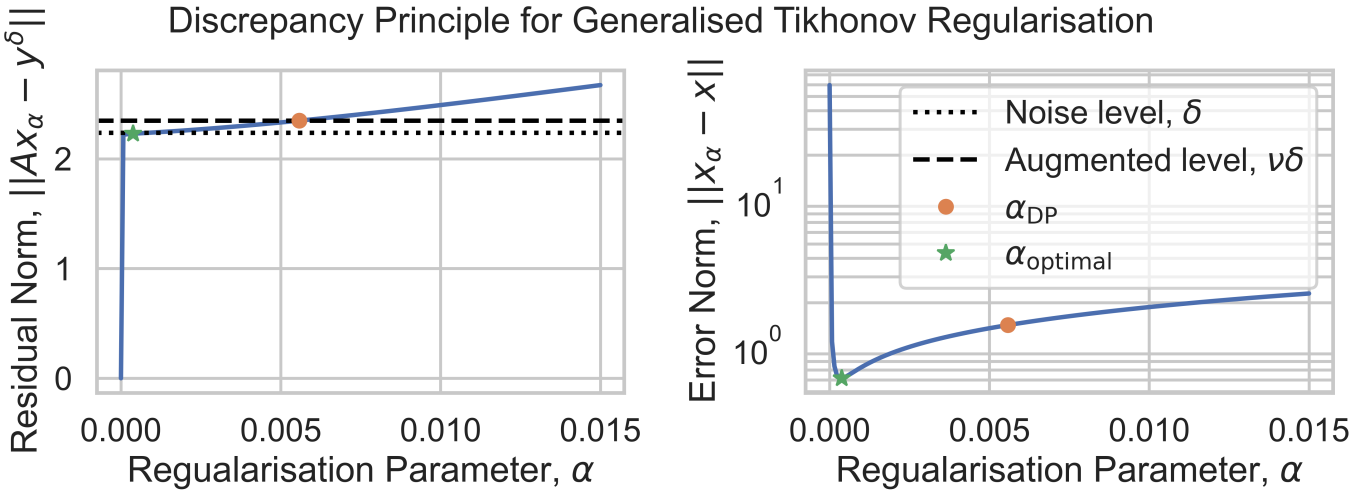
In previous slide we had $D = 0$, i.e. ordinary Tikhonov.

Now consider generalised Tikhonov with D as central difference matrix.

Note: Null-space of D are all constant vectors $\Rightarrow$ Derivative term does not penalise/bias constant offsets

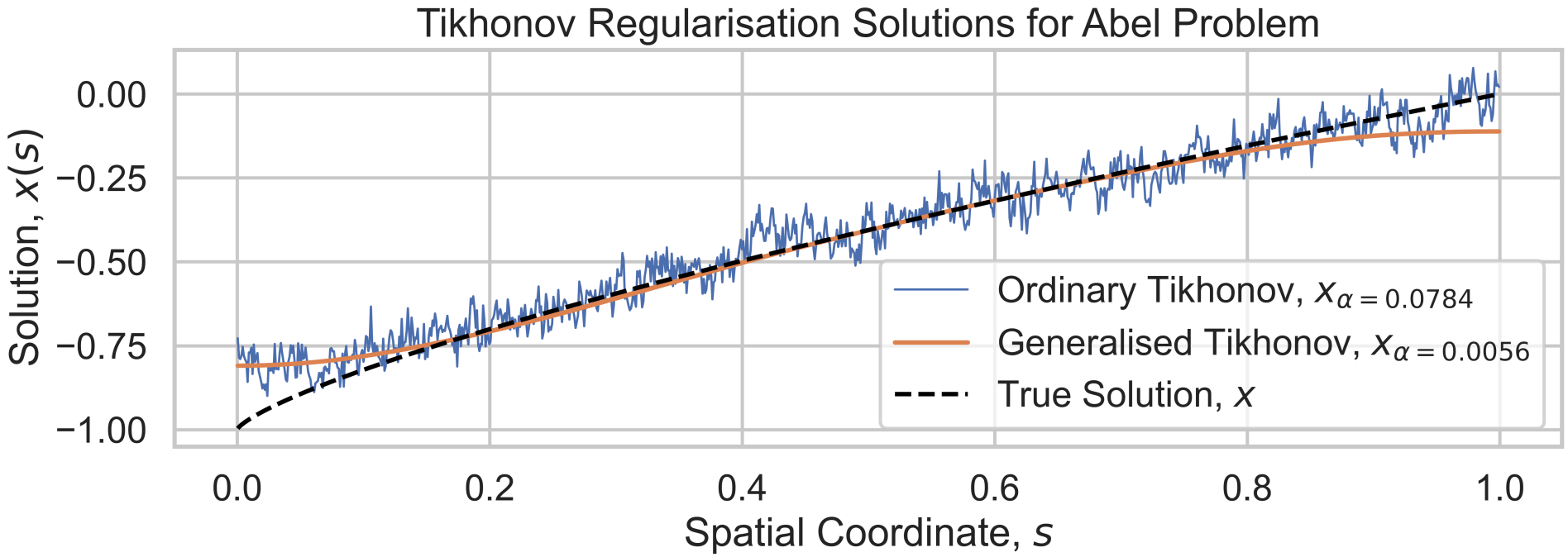
$$D = \frac{1}{h}\begin{pmatrix} -1 & 1 & & & \\ -\frac{1}{2} & 0 & \frac{1}{2} & & \\ & \ddots & \ddots & \ddots & \\ & & -\frac{1}{2} & 0 & \frac{1}{2} \\ & & & -1 & 1 \end{pmatrix},$$

$$h = \frac{1}{N}$$



- Overshoots optimal value
- Lower errors possible than with ordinary Tikhonov
  - Better able to exploit smoothness of true solution

Abel Problem — Reconstruction Results



- Over-regularisation apparent in reconstruction
- Generalised Tikhonov able to attenuate HF noise using derivative term